

Gabriele Fazio

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ABOUT ME

I am a Physics Master's student, published researcher, and an Italian & Swiss citizen with a dedicated specialization in quantum technologies. I stay actively engaged with the field by regularly attending the quantum seminars held at my university. I am seeking a PhD position in Quantum Computing where I can apply my robust theoretical foundation to complex challenges.

EDUCATION

Università degli Studi di Milano Master's Degree in Physics – Master's Thesis: <i>Block-encoding techniques for quantum computing applications</i>	Milan, Italy [September 2024 – Expected Graduation: September 2026]
Bachelor's Degree in Physics – Bachelor's Thesis: <i>Cartan quantum metrology</i> – Graduated with 105/110	[October 2021 – July 2024]
Liceo Scientifico Statale Orazio Grassi High school diploma – Graduated with 100/100	Savona, Italy [September 2016 – June 2021]
Università degli Studi di Genova Cybersecurity training (<i>Cyberchallenge program</i>)	Genoa, Italy [February 2019 – May 2019]

PUBLICATIONS

Peer-Reviewed Journal Articles

- **Fazio G**, He J and Paris M G A (2026). "Order matters: tight bounds on the precision of sequential quantum estimation for multiparameter models" *Journal of Physics A: Mathematical and Theoretical*, 59(4), 045304. [\[DOI\]](#)
- **Fazio G**, He J and Paris M G A (2025). "Cartan quantum metrology" *Physics Open*, 23, 100260. [\[DOI\]](#)

Preprints

- He J, **Fazio G** and Paris M G A (2025). "Weight-dependent and weight-independent measures of quantum incompatibility in multiparameter estimation" *arXiv*, 2510.18864. [\[arXiv\]](#)

RESEARCH EXPERIENCE

Master's Research:

Supervised by **Prof. Enrico Prati** & **Sebastiano Corli**

- **Focus:** Investigating the implementation of **Block Encoding** techniques and **QRAM** (Quantum Random Access Memory) architectures to execute linear algebra routines on quantum hardware for **Neural Network** models.
- **Interdisciplinary Tools:** Utilizing **Random Matrix Theory** to model **singular value distributions** of neural network layers to properly characterize the scaling of **Circuit Complexities**.
- **Key Skills:** Advanced algorithms knowledge, Python/PennyLane simulations, and high-level mathematical modeling.

Quantum Metrology Research:

Research Assistant, Supervised by **Prof. Matteo Paris**

- **Project:** Comprehensive study of the landscape of bounds (Cramér-Rao, Holevo, ...) in **Quantum Multiparameter Metrology**.
- **Key Skills:** Mastered the transition from theoretical derivation to publication-ready manuscripts.

TECHNICAL SKILLS

Programming Languages: Python, Mathematica, C++, C, Bash, JavaScript, SQL

Quantum & Scientific Tools: PennyLane, Qiskit, NumPy, TensorFlow, Keras, PyTorch, scikit-learn, Matplotlib, Jupyter

Software & Environments: Git/GitHub, Unix, LaTeX, Microsoft 365 Suite

CONTACT DETAILS OF REFERENCES

- **Enrico Prati** - enrico.prati@unimi.it: Master's thesis supervisor.
- **Matteo G. A. Paris** - matteo.paris@fisica.unimi.it: Bachelor's thesis supervisor and co-author of 3 papers.

LANGUAGES

Italian: Native | **English:** Advanced | **French:** Elementary

AWARDS & COMPETITIONS

- **Bronze medal** in Italian Cybersecurity Olympiads, [Edition 2021](#).
- **National Finalist - OIS:** Ranked 1st in the region to represent Liguria at the [OIS](#) (Olimpiadi di Informatica a Squadre) National Finals in both 2018/19 and 2019/20 editions.

COMMUNITY OUTREACH

Giovani per la Scienza (Cultural association)

Member & group leader

Savona, Italy

[September 2017 - April 2019]

- **Leadership:** Promoted to Group Leader in the second year, coordinating a team to design and execute technical projects.
- **Scientific Outreach:** Presented live physics experiments at public exhibitions, simplifying complex physical phenomena for diverse audiences to promote scientific literacy.
- **Active Membership:** Participated as a regular member, attending weekly physics lessons and collaborative group activities.

